

Lecture 9: Unbiasedness, Rao-Blackwell Theorem.

Definition

$X_1, \dots, X_n$ is a random sample from a distribution $P_\theta \in \mathcal{P}$, $\theta \in \Theta$.

$\delta(\underline{x})$ is an unbiased estimator for $g(\theta)$ if

$$E(\delta(\underline{x})) = g(\theta) \quad \forall \theta \in \Theta.$$

Ex 1: $X_1, \dots, X_n$ iid from $N(\theta, 1)$, $\theta \in \mathbb{R}$.

The estimand $g(\theta) = \theta$, then

$$\delta_1(\underline{x}) = x_1, \quad \delta_2(\underline{x}) = \frac{1}{2}(x_1 + x_2), \quad \delta_3(\underline{x}) = \bar{x}, \quad \delta_4(\underline{x}) = \sum_{i=1}^n a_i x_i, \quad \sum a_i = 1$$

all are u.e.'s of $g(\theta)$.

Ex 2 $X_1, \dots, X_n$ iid from $U(0, \theta)$, $\theta > 0$, $g(\theta) = \theta$

$$\delta_1(\underline{x}) = 2x_1, \quad \delta_2(\underline{x}) = x_1 + x_3, \quad \delta_3(\underline{x}) = 2\bar{x}, \quad \delta_4(\underline{x}) = \frac{n+1}{n} X_{(n)}$$

are all u.e.'s of $g(\theta) = \theta$.

Ex 3 $X_1, \dots, X_n$ iid from $\text{Ber}(\theta)$, $\theta \in (0, 1)$.

(a) $g(\theta) = \theta$, then

$$\delta_1(\underline{x}) = x_i \quad \forall i=1(1)n, \quad \delta_2(\underline{x}) = \frac{x_1 + x_5 + x_8}{3}, \quad \delta_3(\underline{x}) = \frac{1}{n} \sum x_i \text{ are some u.e.'s of } g(\theta)$$

(b) Let $g(\theta) = \theta(1-\theta)$, to get an u.e. of $g(\theta)$, define

$$\delta(\underline{x}) = \begin{cases} 1 & \text{if } x_1=1, x_2=0 \\ 0 & \text{or} \end{cases}$$

$$\text{Then } E\delta(\underline{x}) = P(x_1=1, x_2=0) = P(x_1=1)P(x_2=0) = \theta(1-\theta) \quad \forall \theta$$

so that $\delta(\underline{x})$ defined above is an u.e. of $g(\theta) = \theta(1-\theta)$.

Ex 4 An unbiased estimator may not exist.

Single obs_n X from $\text{Bin}(n, \theta)$, $\theta \in (0, 1)$; $g(\theta) = \frac{1}{\theta}$.

Then a $\delta(\underline{x}) \ni E(\delta(\underline{x})) = \frac{1}{\theta} \quad \forall \theta \in (0, 1)$ does not exist!

Why?

Thus, for a parameter $g(\theta)$, there can be more than one, even infinite, unbiased estimators. We need to pick up the 'best' one, i.e., one having the minimum variance $\forall \theta$, i.e., to find $\delta^*(\underline{x}) \ni$

$$(i) E(\delta^*(\underline{x})) = g(\theta) \quad \forall \theta \in \Theta$$

$$(ii) \text{Var}(\delta^*(\underline{x})) \leq \text{Var}(\delta(\underline{x})) \quad \forall \theta \in \Theta, \quad \forall \delta \text{ satisfying (i).}$$

Such a $\delta^*(\underline{x})$ is the Uniformly Minimum Variance Unbiased Estimator (UMVUE)

Improvement of an unbiased estimator using a sufficient statistic, or, Rao-Blackwellization of an unbiased estimator.

The following theorem is a stepping stone towards achieving the UMVUE, but does not immediately give the UMVUE.

It provides a way to improve upon an unbiased estimator, in terms of lower variance.

Rao-Blackwell Theorem

(In the usual set up) Let $\delta(\underline{x})$ be any unbiased estimator of $g(\theta)$
 $T(\underline{x})$ be a sufficient statistic for θ

Define $\eta(T) = E(\delta(\underline{x}) | T)$, then

- || (i) $\eta(T)$ is a statistic
 (ii) $\eta(T)$ is an unbiased estimator of $g(\theta)$ (like $\delta(\underline{x})$)
 (iii) $\text{Var}(\eta(T)) \leq \text{Var}(\delta(\underline{x}))$, '=' iff $\eta(T) = \delta(\underline{x})$ w.p. 1.

Pf (i) $\because T$ is suff., the conditional distr $P(\underline{x}_1, \dots, \underline{x}_n | T=t)$ does not depend on θ . Hence $E(\delta(\underline{x}) | T)$ does not depend on θ , and is a statistic.

$$(ii) E_T(\eta(T)) = E_T[E_{\underline{x}|T}(\delta(\underline{x}) | T)] = E_{\underline{x}} \delta(\underline{x}) = g(\theta) \quad \forall \theta \in \Theta$$

$$(iii) \text{Var}(\eta(T)) = \text{Var}[E(\delta(\underline{x}) | T)] \\ = \text{Var}(\delta(\underline{x})) - E[\text{Var}(\delta(\underline{x}) | T)] \\ \leq \text{Var}(\delta(\underline{x}))$$

(Using, law of total variance, for r.v.s Y and Z , $\text{Var}(Y) = \text{Var}(E(Y|Z)) + E(\text{Var}(Y|Z))$.
 To prove this, $\text{Var}(Y) = E(Y - E(Y))^2 = E(Y - E(Y|Z) + E(Y|Z) - E(Y))^2$,
 and proceed)

The above method, popularly called Rao-Blackwellization leads to a new estimator $\eta(T)$, which is unbiased like the original unbiased u.e. $\delta(\underline{x})$, but has a lower variance than $\delta(\underline{x})$.

All we need is a sufficient statistic to facilitate this improvement.

Ex 1 $X_1, \dots, X_n$ r.s. from Bin. $(1, \theta)$ (or, $\text{Ber}(\theta)$), $0 < \theta < 1$, $g(\theta) = \theta$ 3/9
 $T = \sum_{i=1}^n X_i$ is sufficient (in fact, minimal sufficient)

$\delta(\underline{x}) = X_1$ is an unb. est. of $g(\theta) = \theta$.

Then $\eta(T) = E(\delta(\underline{x}) | T)$

$$\begin{aligned} \text{so that } \eta(t) &= P(X_1 = 1 | T=t) \quad (\because X_1 = \begin{cases} 1 & \text{w.p. } \theta \\ 0 & \text{w.p. } 1-\theta \end{cases}) \\ &= \frac{P(X_1=1, \sum_{i=1}^n X_i = t)}{P(\sum_{i=1}^n X_i = t)} = \frac{P(X_1=1, \sum_{i=2}^n X_i = t-1)}{P(\sum_{i=1}^n X_i = t)} = \frac{P(X_1=1) \cdot P(\sum_{i=2}^n X_i = t-1)}{P(\sum_{i=1}^n X_i = t)} \\ &= \frac{\theta(1-\theta)^{n-1} \cdot \theta^{t-1} (1-\theta)^{n-1-t}}{\binom{n}{t} \theta^t (1-\theta)^{n-t}} = \frac{\binom{n-1}{t-1}}{\binom{n}{t}} = \frac{t}{n} \end{aligned}$$

so that $\eta(T) = \frac{T}{n}$, Note $E(\eta(T)) = \frac{n\theta}{n} = \theta = E(X_1) = E(\delta(\underline{x})) \forall \theta$
 but $\text{Var}(\eta(T)) = \frac{n\theta(1-\theta)}{n^2} = \frac{\theta(1-\theta)}{n} \leq \text{Var}(X_1) = \theta(1-\theta)$

'=' iff $n=1$, i.e. $T = X_1 (= \delta(\underline{x}))$ w.p. 1

Ex 2 $X_1, \dots, X_n$ r.s. from $P(\theta)$, $\theta > 0$
 $g(\theta) = e^{-\theta}$
 $T(\underline{x}) = \sum_{i=1}^n X_i$ is sufficient for θ .

Note, $\delta(\underline{x}) = \begin{cases} 1 & \text{if } X_1 = 0 \\ 0 & \text{or} \end{cases}$ is unb. for $g(\theta) = e^{-\theta}$.

$$\begin{aligned} \text{Then } \eta(T) &= E(\delta(\underline{x}) | T) \\ \eta(t) &= E(\delta(\underline{x}) | T=t) = P(X_1=0 | T=t) = \frac{P(X_1=0, T=t)}{P(T=t)} = \frac{P(X_1=0, \sum_{i=2}^n X_i = t)}{P(\sum_{i=1}^n X_i = t)} \\ &= \frac{P(X_1=0) P(\sum_{i=2}^n X_i = t)}{P(\sum_{i=1}^n X_i = t)} = \frac{e^{-\theta} \cdot e^{-(n-1)\theta} [(n-1)\theta]^t / t!}{e^{-n\theta} (n\theta)^t / t!} = \left(\frac{n-1}{n}\right)^t \end{aligned}$$

Hence, $\eta(T) = \left(\frac{n-1}{n}\right)^T$, $n \geq 2$ with $E(\eta(T)) = g(\theta) = e^{-\theta}$
 and $\text{Var}(\eta(T)) \leq \text{Var}(\delta(\underline{x}))$, $\forall \theta > 0$.

In both the above examples, Rao-Blackwellization has been done with a complete, sufficient statistic, rather than just any sufficient (or, minimal sufficient statistic). This has led us to the unique UMVUE $\eta(T)$ in both the cases!

Next, completeness of a statistic